AAI- 530-IN2

Data Analytics and Internet of Things

FINAL TEAM PROJECT
GROUP 6

University of San Diego
SHILEY-MARCOS SCHOOL OF ENGINEERING

Smart Building IoT System

Energy and Environmental Monitoring with Machine Learning Analytics

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Course: IoT Systems and Machine Learning

Project Overview

This project implements a complete IoT system for smart building monitoring, combining real-world sensor data analysis with advanced machine learning techniques. The system monitors energy consumption, temperature, humidity, and occupancy across multiple zones in a university building infrastructure.

Key Features

- Complete IoT system architecture design with edge computing and cloud processing
- Deep learning LSTM network for time series prediction
- CNN-based occupancy classification for energy optimization
- Interactive Tableau dashboard for real-time monitoring and insights
- Comprehensive data processing pipeline with exploratory analysis
- Production-ready code with modular architecture

Dataset

Source: Zenodo - Smart Building IoT Dataset

Citation: Aboubakar, A., Saeed, N., & Zaki, N. (2024). Dataset of IoT-based energy and environmental parameters in a smart building infrastructure. *Data in Brief*, 55, 110672.

Dataset Characteristics:

- **Collection Period:** January 1, 2024 - June 20, 2024 (171 days)
- **Total Records:** 123,000+ timestamped measurements
- **Sensor Types:** Energy meters (smart plugs), Temperature/Humidity (DHT22), Occupancy (PIR)
- **Locations:** Interdisciplinary Lab, Kitchen, Mail Room, Office Zones
- **Appliances Monitored:** Coffee machines, Microwaves, Refrigerators, HVAC systems

Why This Dataset:

1. Real production IoT infrastructure (not simulated)
2. Multiple time series variables for comprehensive analysis
3. Sufficient data volume for deep learning training
4. Practical application with energy efficiency implications

Installation Process

Requirements

- Python 3.9 or higher
- TensorFlow 2.13+
- GitHub

Setup Instructions

1. **Clone the repository:**
2. **Create virtual environment:**
3. **Install dependencies:**
4. **Download dataset:**
    a. Visit Zenodo dataset page
    b. Download CSV files
    c. Place in `data/raw/` directory

Usage

1. Data Exploration

Run the portion of the notebook to explore the dataset:

- Loading and inspecting data
- Missing value analysis
- Temporal patterns visualization
- Statistical summary
- Correlation analysis

2. Data Preprocessing

Process and engineer features:

- Data cleaning and outlier handling
- Temporal feature extraction
- Lag and rolling statistics
- Domain-specific features
- Data normalization

3. LSTM Temperature Prediction

Train deep learning model for temperature forecasting:

- Sequence creation for LSTM
- Model architecture design
- Training with callbacks
- Multi-horizon evaluation (1h, 3h, 6h)
- Prediction visualization

4. CNN Occupancy Classification

Train CNN for occupancy state prediction:

- Label generation from sensor patterns
- Windowed data preparation
- CNN architecture design
- Multi-class classification training
- Confusion matrix analysis

IoT System Design

The project includes a complete theoretical IoT system design covering:

System Components

1. **Perception Layer**
    a. Energy monitoring sensors (Shelly EM smart plugs)
    b. Environmental sensors (DHT22 temperature/humidity)
    c. Occupancy sensors (PIR motion detectors)
2. **Edge Computing**
    a. Raspberry Pi 4 edge devices (3 units)
    b. Local data aggregation and buffering
    c. Anomaly detection at edge
    d. MQTT broker (Eclipse Mosquitto)
3. **Network Architecture**
    a. WiFi 802.11ax (WiFi 6) connectivity
    b. Isolated IoT VLAN
    c. TLS 1.2 encryption
    d. MQTT over secure channels
4. **Cloud Infrastructure (AWS)**

a. AWS IoT Core for device management
b. Amazon Timestream for time series storage
c. AWS Lambda for serverless processing
d. Amazon SageMaker for ML training

5. **Application Layer**
   a. Interactive Tableau dashboard
   b. REST API for data access
   c. Real-time alerts and notifications

Dashboard

Tableau Public Dashboard:

View Live Dashboard 1: SummaryofOverallEnergyConsumptionTrends

View Live Dashboard 2: DailyHourlyCoffeeMachineUsagePatterns

View Live Dashboard 3: LSTMModelPerformanceMonitoring-CoffeeMachineKWh

Dashboard Components

Status Visualizations.

Summary Visualizations.

ML Insights Visualizations.

Energy Savings Potential

- **Occupancy-based HVAC:** 23,400 kWh/year ($3,510 savings)
- **Predictive temperature control:** 16,640 kWh/year ($2,496 savings)
- **Equipment optimization:** 9,360 kWh/year ($1,404 savings)
- **Total projected savings:** 38% energy reduction, $7,410/year

Technologies Used

Programming & Data Science:

- Python 3.9+
- Pandas, NumPy, SciPy
- Matplotlib, Seaborn, Plotly

Machine Learning & Deep Learning:

- TensorFlow 2.13
- Keras API
- Scikit-learn

IoT & Cloud (Reference):

- MQTT Protocol
- AWS IoT Core
- Amazon Timestream
- InfluxDB

Visualization:

- Tableau Public
- Jupyter Notebooks

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Citation

Dataset

- Al-Dujaili, A. (2024). *Dataset of IoT-Based Energy and Environmental Parameters in a Smart Building Infrastructure 2.0* [Data set]. Zenodo. [redacted link]

Software and Libraries

- **Google Colaboratory**: Google. (n.d.). *Google Colaboratory*.
- **NumPy**: Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., ... & Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357-362.
- **Pandas**: McKinney, W. (2010). Data Structures for Statistical Computing in Python. In S. van der Walt & J. Millman (Eds.), *Proceedings of the 9th Python in Science Conference* (Vol. 445, pp. 51-56).
- **Matplotlib**: Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90-95.
- **Scikit-learn**: Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, *12*, 2825-2830.
- **Seaborn**: Waskom, M. L. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, *6*(60), 3021.
- **PyTorch**: Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, *32*.
- **Tableau**: Tableau Software. (n.d.). *Tableau Desktop* [Computer software].

Additional Resources

- Zenodo Dataset Repository
- Project Documentation
- IoT System Design
- Tableau Dashboard